Embedded Linux & System Software Interview Handbook

Chapter 4 - Board Bring-up (Part 5)

Topics: UART Log Analysis, JTAG, GDB, Boot Time Optimization, Real Debugging Case Studies

4.21 Reading UART Boot Logs

Experienced embedded engineers determine the failing stage by examining the last UART message. Never read logs randomly—identify the final successful step.

Example Boot Log

ROM: Booting...
Loading SPL...
DDR initialized
Loading U-Boot...
U-Boot 2025.01
Loading Kernel...
Starting kernel...

If the log stops at **Starting kernel...**, the issue is usually after U-Boot has completed successfully. Focus on the kernel image, Device Tree, boot arguments, DDR stability or early kernel initialization.

4.22 Boot Time Measurement

Reducing boot time is an important optimization task in TVs, automotive systems and IoT devices.

Stage	Typical Optimization
BootROM	Not configurable
SPL	Reduce hardware initialization
U-Boot	Disable unnecessary commands
Kernel	Build only required drivers
Userspace	Delay non-critical services

Useful Commands

• dmesg
• systemd-analyze
• systemd-analyze blame
• printenv (U-Boot)
• bdinfo (U-Boot)

4.23 JTAG Debugging

JTAG provides hardware-level debugging even before UART is functional. It allows engineers to halt the CPU, inspect registers, single-step instructions and download firmware directly into RAM.

When to use JTAG

• No UART output
• BootROM crashes
• CPU stuck in reset
• Early exception handling
• Secure boot investigation

4.24 GDB Debugging

GDB is commonly used with OpenOCD or vendor probes. Typical workflow:

Target Board ↔ JTAG Probe ↔ OpenOCD ↔ GDB

Useful commands:
target remote :3333
break start_kernel
continue
info registers
x/16x 0xADDRESS

4.25 Real Bring-up Case Study

Problem: New board shows no UART output.

Investigation:
1. Measure CPU core voltage.
2. Verify PMIC Power-Good signals.
3. Check reset pin using oscilloscope.
4. Verify oscillator frequency.
5. Confirm boot mode straps.
6. Attach JTAG and inspect Program Counter.
7. Validate UART pin multiplexing.

Root Cause Example: UART TX pin configured for GPIO instead of UART in early firmware.

Senior Interview Scenario

Question: 'A board boots to U-Boot but hangs before the Linux login prompt. How would you debug it?'

A strong answer covers: bootargs, DTB, root filesystem, storage driver, kernel logs, init process, memory corruption, and reproducing the issue on a known-good image.

Common Mistakes

• Blaming Linux before validating hardware.
• Ignoring power sequencing.
• Not checking UART baud rate.
• Using the wrong Device Tree.
• Forgetting earlycon during kernel debugging.
• Not comparing against a known working board.

Key Takeaways

Successful board bring-up depends on structured debugging. Always isolate the failing boot stage, collect evidence from UART/JTAG, validate hardware fundamentals, and then investigate software. Senior interviewers value this systematic approach more than memorized commands.